CM5 socket, M.2 connector

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

AC power, DC power, AC sync

File: ac_dc_power.kicad_sch

Isolated digital IO, Fan controller, Encoder

File: digital_io_fan_encoder.kicad_sch

7-port USB3 Hub

File: usb3_hub.kicad_sch

Rear USB, USB-C

File: usb_conn.kicad_sch

Ethernet, Video outputs, Ref clk

File: video_outputs.kicad_sch

DIB 2.0

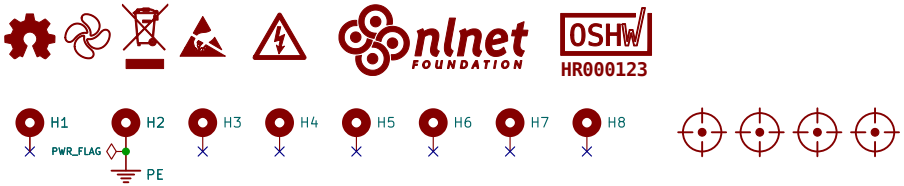
File: dib.kicad_sch

Trigger switch matrix, Power modules coupling

File: power_coupling.kicad_sch

USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub P1 Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

I2C address	Device
0x53	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x3A	PTN5150A USB CC controller
0x70	ADG2188Y 8 x 8 switch array



https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /
File: BB3plus mainboard.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-31	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 1/10

CM5 socket

Since ID_SD and ID_SC pins are used as GPIO pins, then to prevent the firmware from checking to see if there is a HAT EEPROM available, add force_eeprom_read=0 and disable_poe_fan=1 to the config.txt file.

X1

DF40HC(3.0)-100DS-0.4V(51)

X2

DF40HC(3.0)-100DS-0.4V(51)

CM5_PWR_BTN input:

RasPI CM5: Pull low to force power off or power on from previous software powered off state. Internally pulled up to +5V via 10kΩ

STM32H7 CM5: Reset input, internally pulled up to +3V3 via 10kΩ

CM5_BOOT input:

RasPi CM5: A low on this pin forces booting from an RPI server (e.g. PC or a Raspberry PI). If not used leave floating. Internally pulled up via 10kΩ to CM5_3.3V

STM32H7 CM5: BOOT0 input, internally pulled down to GND via 10kΩ

Module1

Raspberry PI CM5 module silkscreen

M.2 connector

X3A

SM3ZS067U410AMR1000

X3B

USB2 Mux

IC1

PI3USB102ZLEX

S=Low: M input
S=High: D input

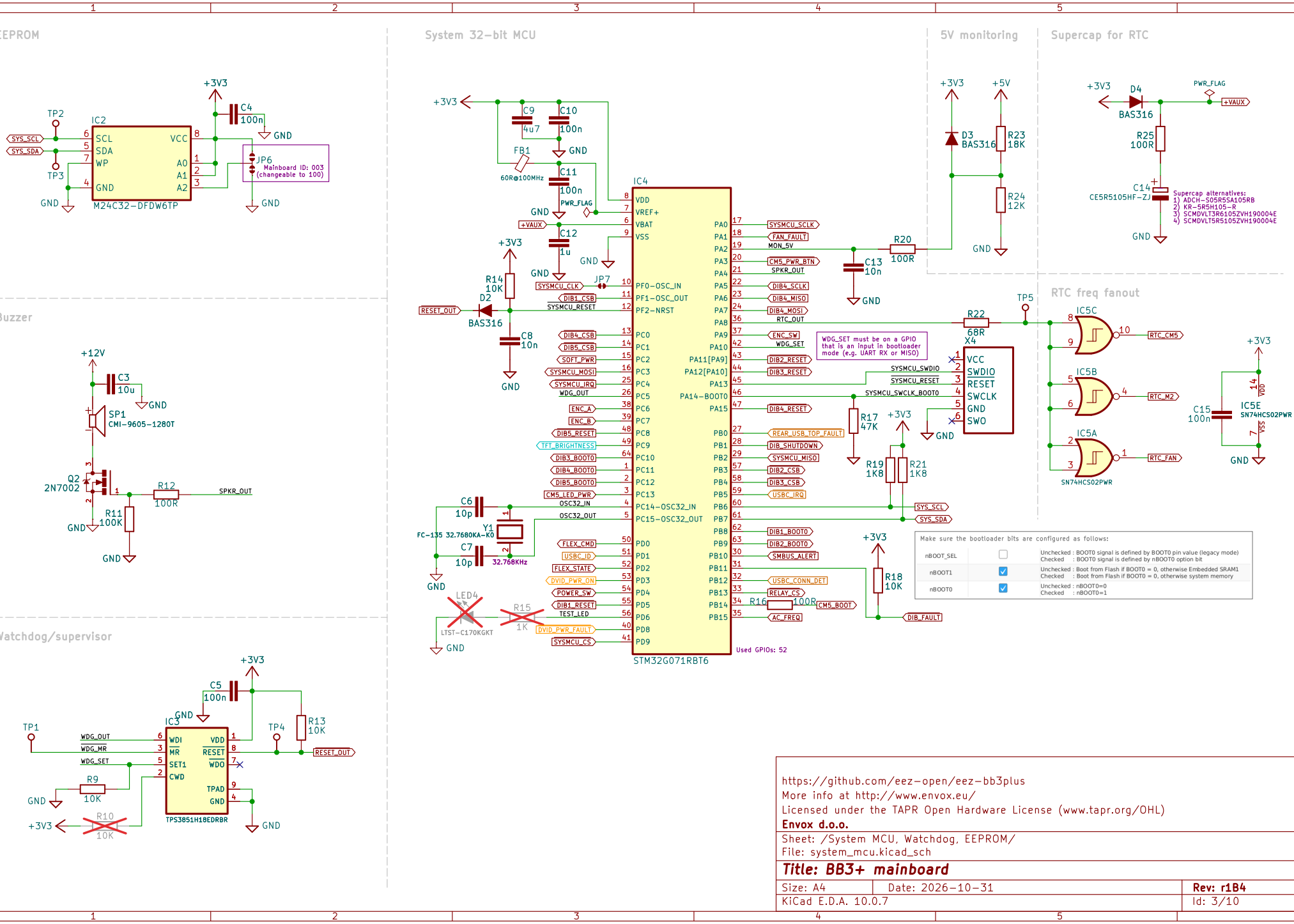
When boot mode is disabled, the USB2 port will be used as a device to access the CM5 via the rpiboot utility. In this state, power must be disabled.

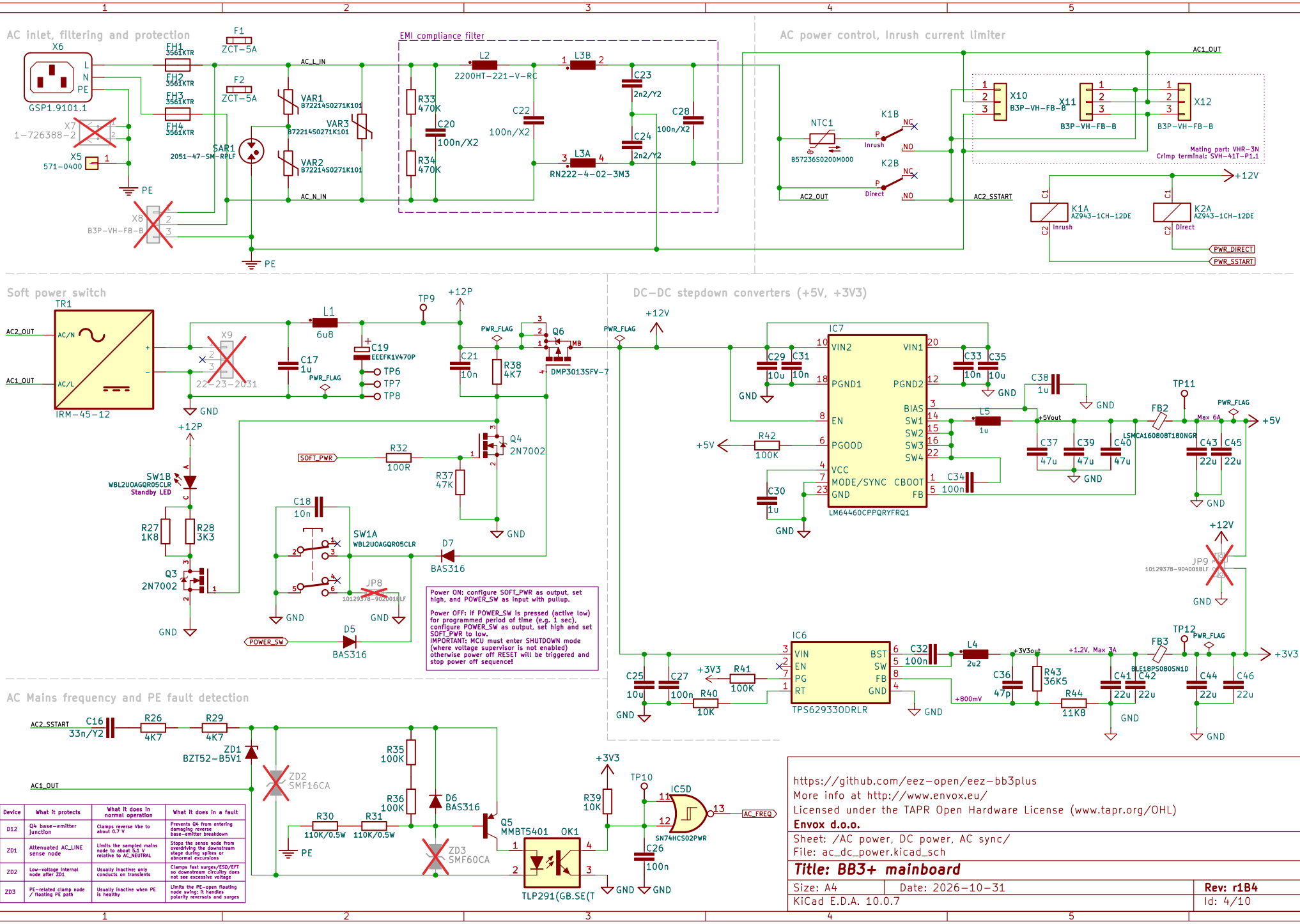
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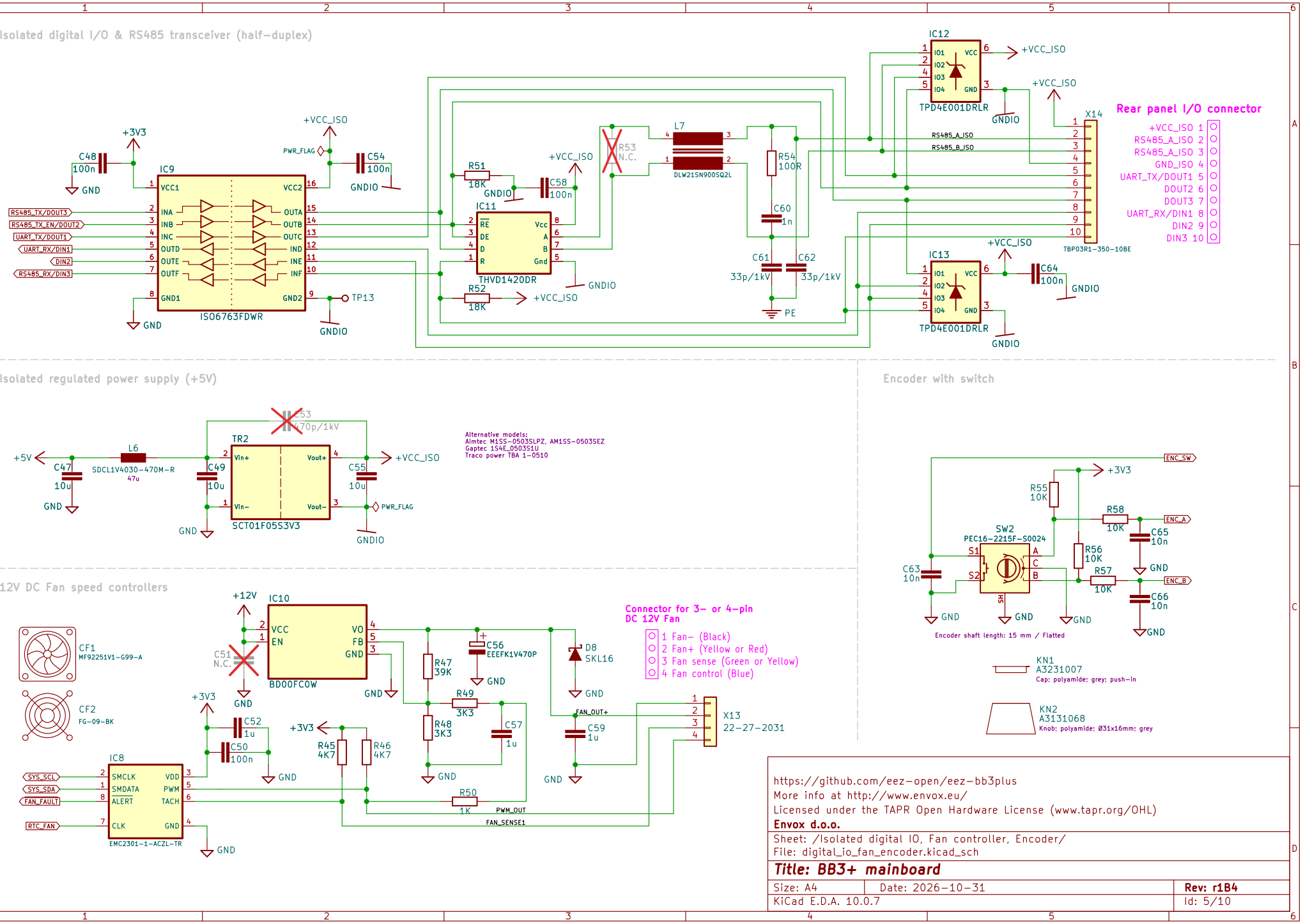
Sheet: /CM5 socket, M.2 connector/
File: cm5_socket.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-31	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 2/10

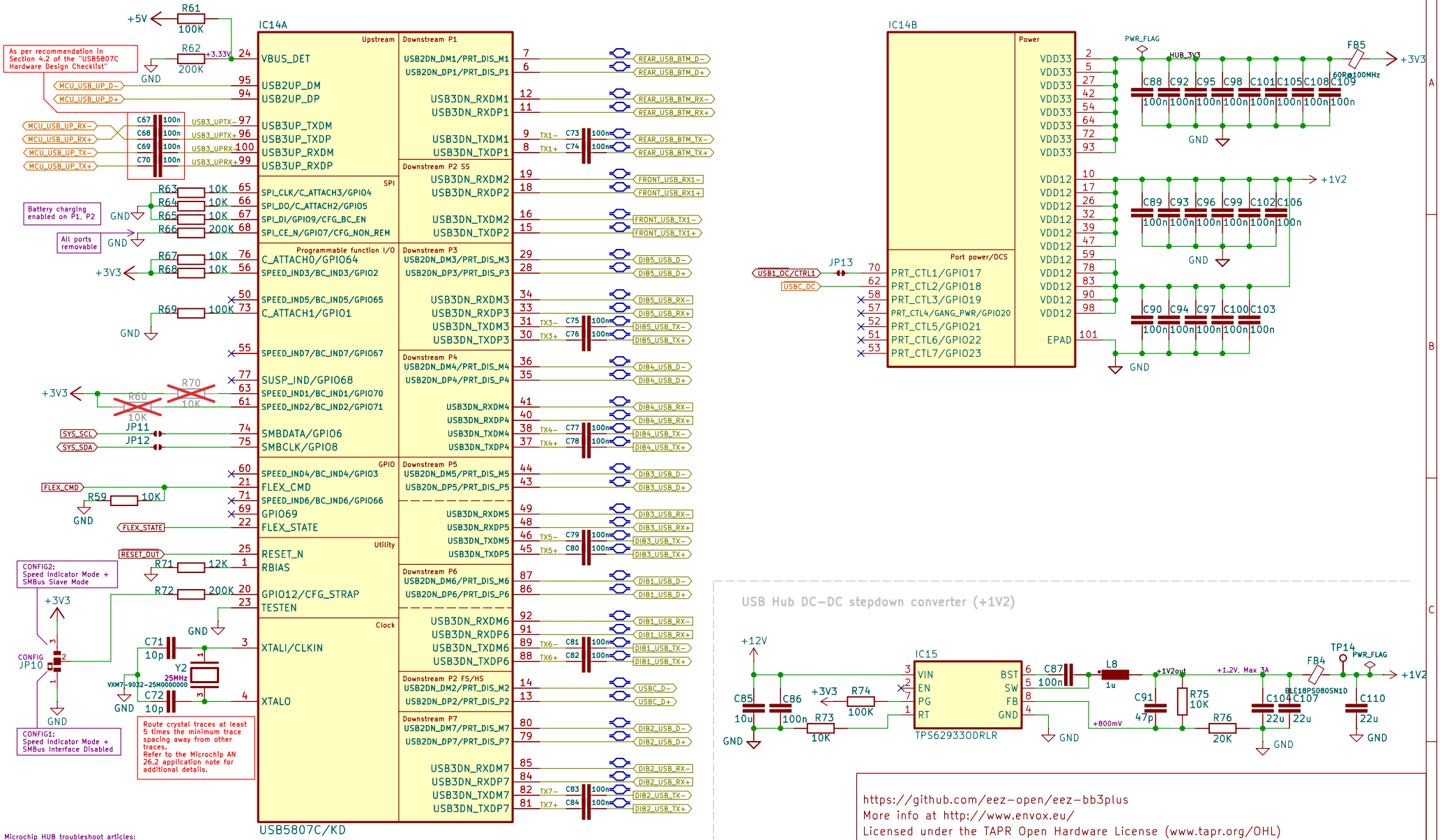

EEPROM
IC2
M24C32-DFDW6TP
TP2
TP3
JP6
Mainboard ID: 003
(changeable to 100)
Buzzer
SP1
CMI-9605-1280T
Q2
2N7002
R12
100R
R11
100K
SPKR_OUT
Watchdog/supervisor
IC3
TPS3851H18EDRBR
TP1
TP4
R13
10K
R9
10K
R10
10K
System 32-bit MCU
IC4
STM32G071RBT6
Used GPIOs: 52
WDG_SET must be on a GPIO that is an input in bootloader mode (e.g. UART RX or MISO)
Make sure the bootloader bits are configured as follows:
nBOOT_SEL
nBOOT1
nBOOT0
Unchecked : BOOT0 signal is defined by BOOT0 pin value (legacy mode)
Checked : BOOT0 signal is defined by nBOOT0 option bit
Unchecked : Boot from Flash if BOOT0 = 0, otherwise Embedded SRAM1
Checked : Boot from Flash if BOOT0 = 0, otherwise system memory
Unchecked : nBOOT0=0
Checked : nBOOT0=1
5V monitoring
D3
BAS316
R23
18K
R24
12K
Supercap for RTC
D4
BAS316
R25
100R
C14
CE5R5105HF-ZJ
Supercap alternatives:
1) ADCH-S05R5SA105RB
2) KR-5R5H105-R
3) SCMDVLT3R6105ZVH190004E
4) SCMDVLT5R5105ZVH190004E
RTC freq fanout
IC5C
IC5B
IC5A
IC5E
SN74HCS02PWR
X4
SWDIO
RESET
SWCLK
GND
SWO
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /System MCU, Watchdog, EEPROM/
File: system_mcu.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-31
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 3/10


AC inlet, filtering and protection
EMI compliance filter
AC power control, Inrush current limiter
Soft power switch
DC–DC stepdown converters (+5V, +3V3)
AC Mains frequency and PE fault detection
X6
GSP1.9101.1
F1
ZCT-5A
F2
ZCT-5A
FH1
FH2
FH3
FH4
3561KTR
VAR1
VAR2
VAR3
B72214S0271K101
SAR1
2051-47-SM-RPLF
X7
1-726388-2
X5
571-0400
X8
B3P-VH-FB-B
PE
AC_L_IN
AC_N_IN
L2
2200HT-221-V-RC
L3A
L3B
RN222-4-02-3M3
R33
470K
R34
470K
C20
100n/X2
C22
100n/X2
C23
2n2/Y2
C24
2n2/Y2
C28
100n/X2
NTC1
B57236S0200M000
K1B
K2B
Inrush
Direct
AC2_OUT
AC2_SSTART
AC1_OUT
X10
X11
X12
B3P-VH-FB-B
Mating part: VHR-3N
Crimp terminal: SVH-41T-P1.1
K1A
K2A
AZ943-1CH-12DE
+12V
PWR_DIRECT
PWR_SSTART
TR1
IRM-45-12
AC/N
AC/L
X9
22-23-2031
L1
6u8
C17
1u
C19
EEEFK1V470P
TP6
TP7
TP8
TP9
+12P
C21
10n
R38
4K7
PWR_FLAG
Q6
DMP3013SFV-7
GND
SW1B
WBL2UOAGQR05CLR
Standby LED
R27
1K8
R28
3K3
Q3
2N7002
SOFT_PWR
R32
100R
Q4
2N7002
R37
47K
C18
10n
SW1A
WBL2UOAGQR05CLR
JP8
10129378-902001BLF
D7
BAS316
D5
BAS316
POWER_SW
Power ON: configure SOFT_PWR as output, set high, and POWER_SW as input with pullup.
Power OFF: if POWER_SW is pressed (active low) for programmed period of time (e.g. 1 sec), configure POWER_SW as output, set high and set SOFT_PWR to low.
IMPORTANT: MCU must enter SHUTDOWN mode (where voltage supervisor is not enabled) otherwise power off RESET will be triggered and stop power off sequence!
IC7
LM64460CPPQRYFRQ1
VIN2
VIN1
PGND1
PGND2
EN
PGOOD
VCC
MODE/SYNC
GND
BIAS
SW1
SW2
SW3
SW4
CBOOT
FB
C29
10u
C31
10n
C33
10n
C35
10u
C38
1u
C30
1u
C34
100n
R42
100K
L5
1u
+5Vout
FB2
LSMCA160808T180NGR
Max 6A
TP11
+5V
C37
47u
C39
47u
C40
47u
C43
22u
C45
22u
JP9
10129378-904001BLF
IC6
TPS629330DRLR
VIN
EN
PG
RT
BST
SW
FB
GND
C25
10u
C27
100n
R41
100K
R40
10K
C32
100n
L4
2u2
+3V3out
+1.2V, Max 3A
FB3
BLE18PS080SN1D
TP12
+3V3
C36
47p
R43
36K5
R44
11K8
+800mV
C41
22u
C42
22u
C44
22u
C46
22u
AC Mains frequency and PE fault detection
AC2_SSTART
C16
33n/Y2
R26
4K7
R29
4K7
ZD1
BZT52-B5V1
ZD2
SMF16CA
AC1_OUT
R35
100K
R36
100K
R30
110K/0.5W
R31
110K/0.5W
D6
BAS316
ZD3
SMF60CA
Q5
MMBT5401
OK1
TLP291(GB.SE(T
R39
10K
TP10
IC5D
SN74HCS02PWR
AC_FREQ
C26
100n
+3V3
GND
Device
What it protects
What it does in normal operation
What it does in a fault
D12
Q4 base–emitter junction
Clamps reverse Vbe to about 0.7 V
Prevents Q4 from entering damaging reverse base–emitter breakdown
ZD1
Attenuated AC_LINE sense node
Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL
Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2
Low-voltage internal node after ZD1
Usually inactive; only conducts on transients
Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3
PE-related clamp node / floating PE path
Usually inactive when PE is healthy
Limits the PE–open floating node swing; it handles polarity reversals and surges
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /AC power, DC power, AC sync/
File: ac_dc_power.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-31
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 4/10


Isolated digital I/O & RS485 transceiver (half-duplex)
IC9
ISO6763FDWR
IC10
BD00FC0W
IC11
THVD1420DR
IC12
TPD4E001DRLR
IC13
TPD4E001DRLR
IC8
EMC2301-1-ACZL-TR
TR2
SCT01F05S3V3
L6
SDCL1V4030-470M-R
L7
DLW21SN900SQ2L
X14
TBP03R1-350-10BE
Rear panel I/O connector
+VCC_ISO 1
RS485_A_ISO 2
RS485_A_ISO 3
GND_ISO 4
UART_TX/DOUT1 5
DOUT2 6
DOUT3 7
UART_RX/DIN1 8
DIN2 9
DIN3 10
Isolated regulated power supply (+5V)
Alternative models:
Aimtec M1SS-0503SLPZ, AM1SS-0503SEZ
Gaptec 1S4E_0503S1U
Traco power TBA 1-0510
Encoder with switch
SW2
PEC16-2215F-S0024
Encoder shaft length: 15 mm / Flatted
KN1
A3231007
Cap; polyamide; grey; push-in
KN2
A3131068
Knob; polyamide; Ø31x16mm; grey
12V DC Fan speed controllers
CF1
MF92251V1-G99-A
CF2
FG-09-BK
Connector for 3- or 4-pin
DC 12V Fan
1 Fan- (Black)
2 Fan+ (Yellow or Red)
3 Fan sense (Green or Yellow)
4 Fan control (Blue)
X13
22-27-2031
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /Isolated digital IO, Fan controller, Encoder/
File: digital_io_fan_encoder.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-31
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 5/10

7-port USB hub

All downstream ports that are not used should be disabled (use I2C to configure).



USB Hub DC-DC stepdown converter (+1V2)

Microchip HUB troubleshoot articles:

https://support.microchip.com/s/article/USB3-Hub-Upstream-Port-Connect-Enumeration-Failure
https://support.microchip.com/s/article/USB-High-Speed-Chirp-Explained
https://support.microchip.com/s/article/USB3-Connection-Process-and-Attach-Detection

IMPORTANT:

If I2C/SMBus pull-up resistors are detected by the USB5807C at start-up, the hub will wait indefinitely to be configured by the attached I2C/SMBus master. For early prototyping, it may be necessary to physically remove the pull-up resistors until the I2C/SMBus master is fully operational and able to properly configure the hub at start-up.

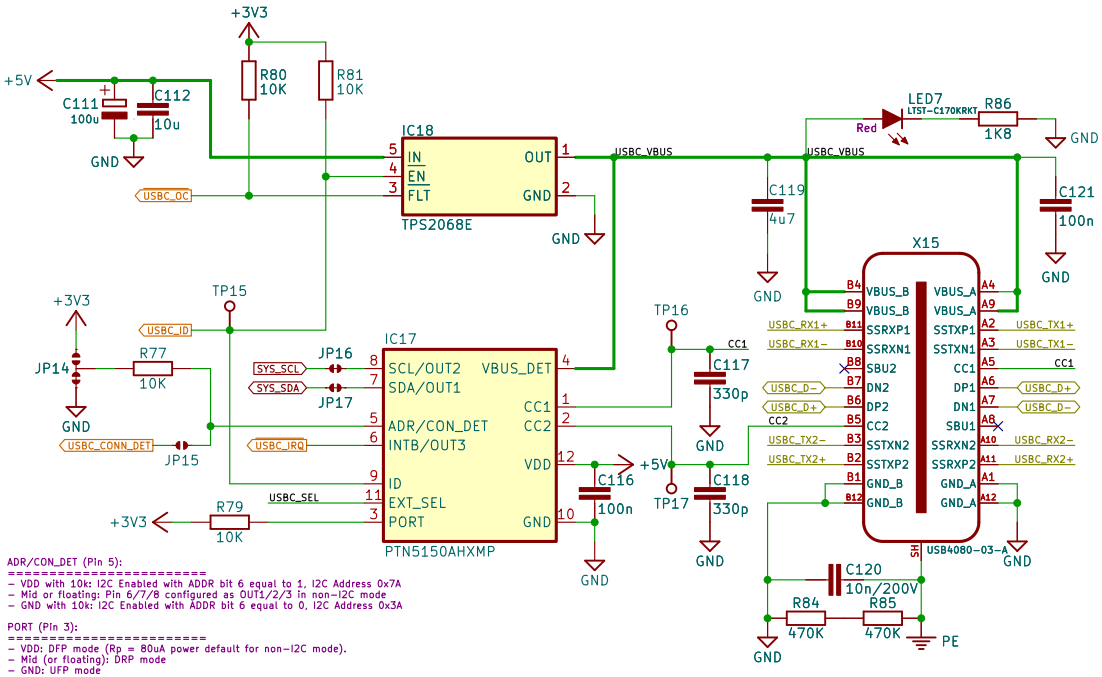
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Sheet: /7-port USB3 Hub/
File: usb3_hub.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-31	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 6/10

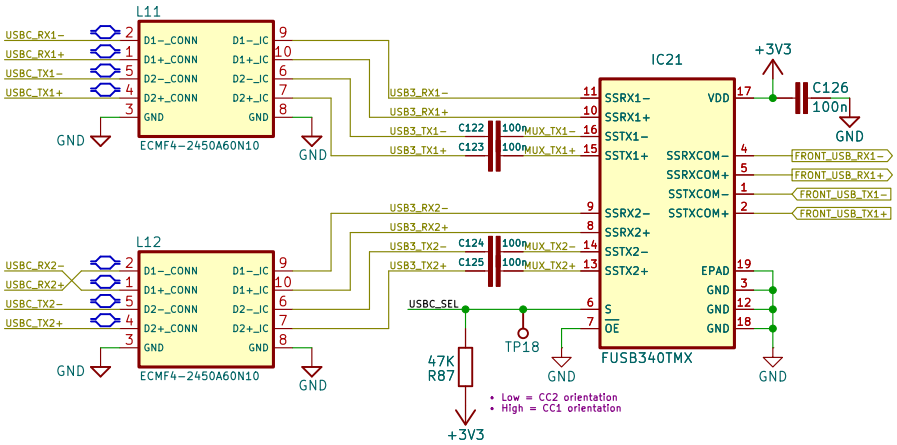
USB-C 3.0 front port (source max 1.5A)



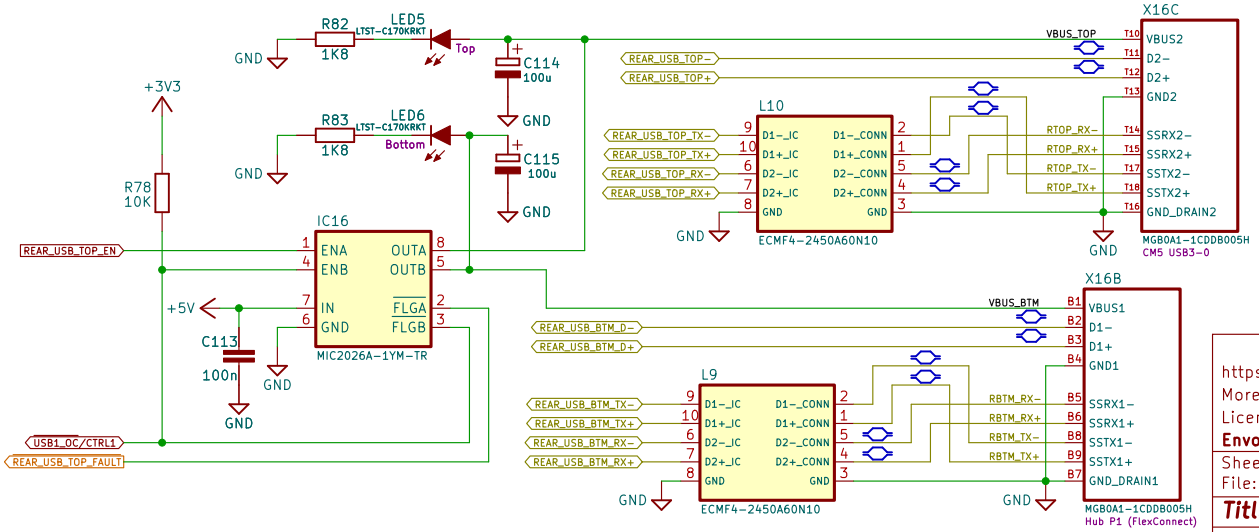
ADR/CON_DET (Pin 5):
- VDD with 10k: I2C Enabled with ADDR bit 6 equal to 1, I2C Address 0x7A
- Mid or floating: Pin 6/7/8 configured as OUT1/2/3 in non-I2C mode
- GND with 10k: I2C Enabled with ADDR bit 6 equal to 0, I2C Address 0x3A

PORT (Pin 3):
- VDD: DFP mode (Rp = 80uA power default for non-I2C mode).
- Mid (or floating): DRP mode
- GND: UFP mode

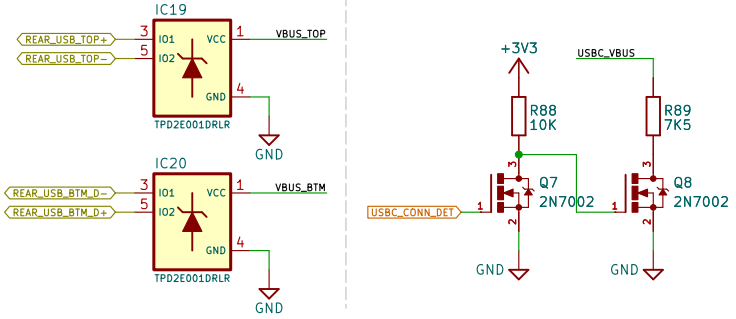
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



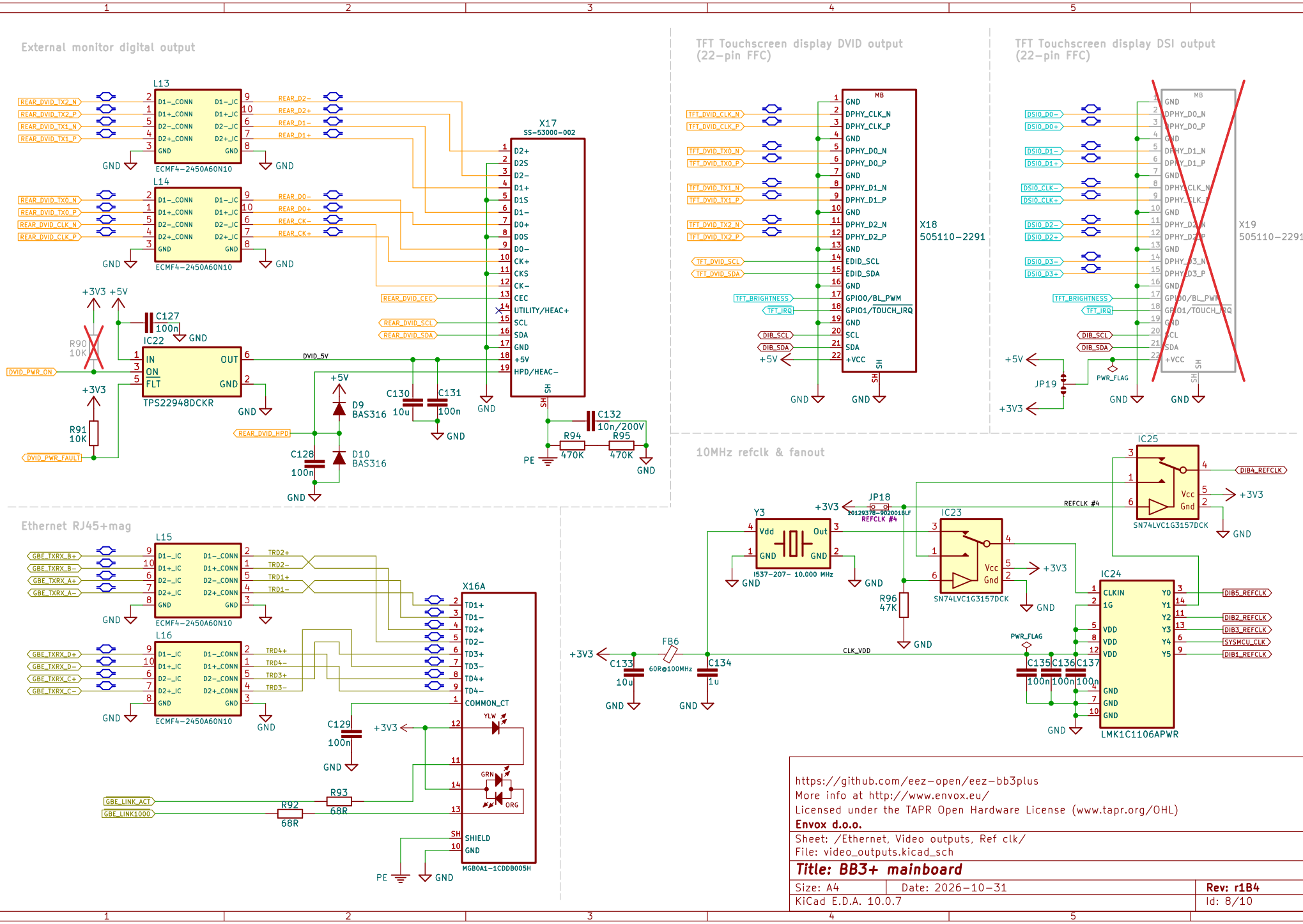
VBus Discharge Circuit

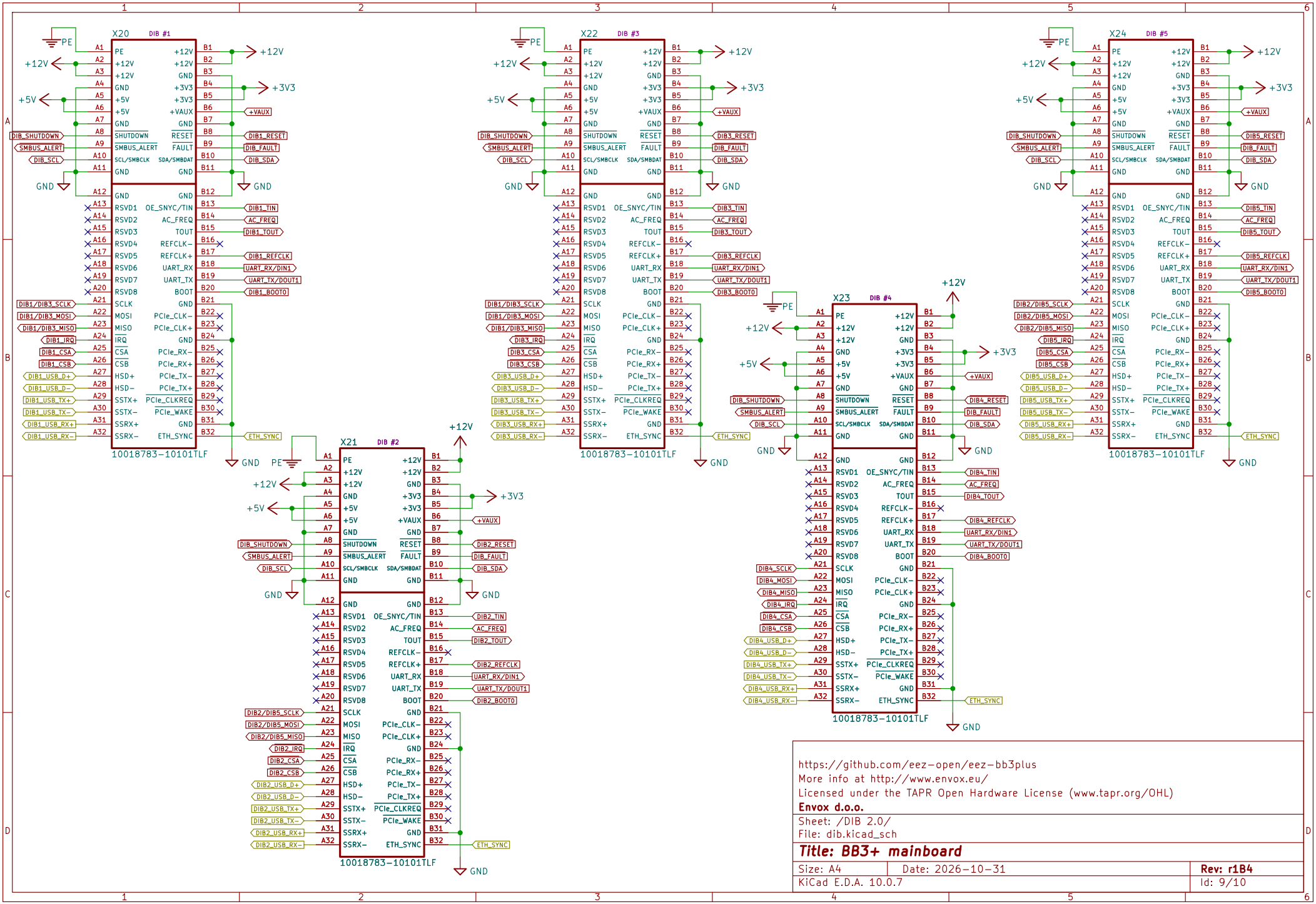


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Sheet: /Rear USB, USB-C/		
File: usb_conn.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-10-31	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 7/10

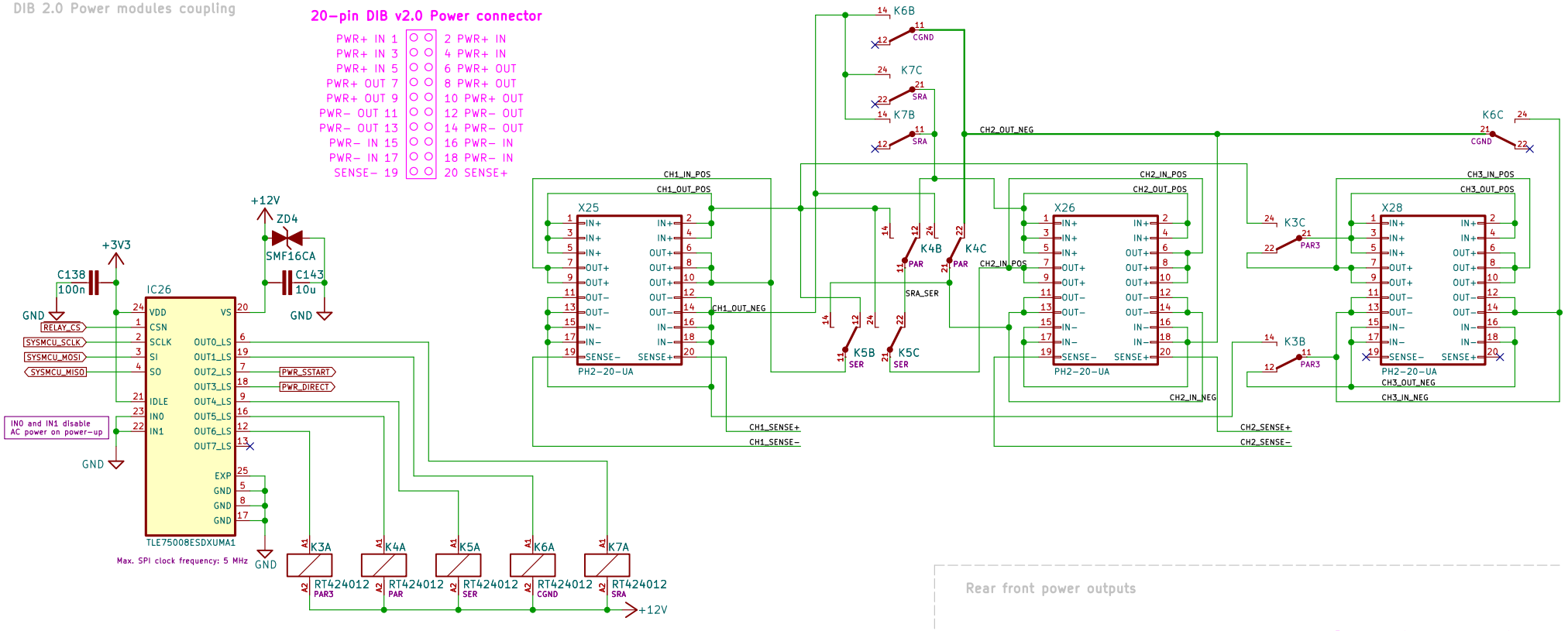

External monitor digital output
L13
ECMF4-2450A60N10
L14
ECMF4-2450A60N10
X17
SS-53000-002
IC22
TPS22948DCKR
C127
100n
R90
10K
R91
10K
C130
10u
C131
100n
C128
100n
D9
BAS316
D10
BAS316
C132
10n/200V
R94
470K
R95
470K
TFT Touchscreen display DVID output
(22-pin FFC)
X18
505110-2291
TFT Touchscreen display DSI output
(22-pin FFC)
X19
505110-2291
JP19
PWR_FLAG
Ethernet RJ45+mag
L15
ECMF4-2450A60N10
L16
ECMF4-2450A60N10
X16A
MGB0A1-1CDDB005H
C129
100n
R92
68R
R93
68R
10MHz refclk & fanout
Y3
I537-207- 10.000 MHz
JP18
2129378-9020018LF
REFCLK #4
IC23
SN74LVC1G3157DCK
IC25
SN74LVC1G3157DCK
R96
47K
FB6
60R@100MHz
C133
10u
C134
1u
IC24
LMK1C1106APWR
C135
100n
C136
100n
C137
100n
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /Ethernet, Video outputs, Ref clk/
File: video_outputs.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-31
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 8/10


X20
DIB #1
X21
DIB #2
X22
DIB #3
X23
DIB #4
X24
DIB #5
10018783-10101TLF
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /DIB 2.0/
File: dib.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-10-31
Rev: r1B4
KiCad E.D.A. 10.0.7
Id: 9/10

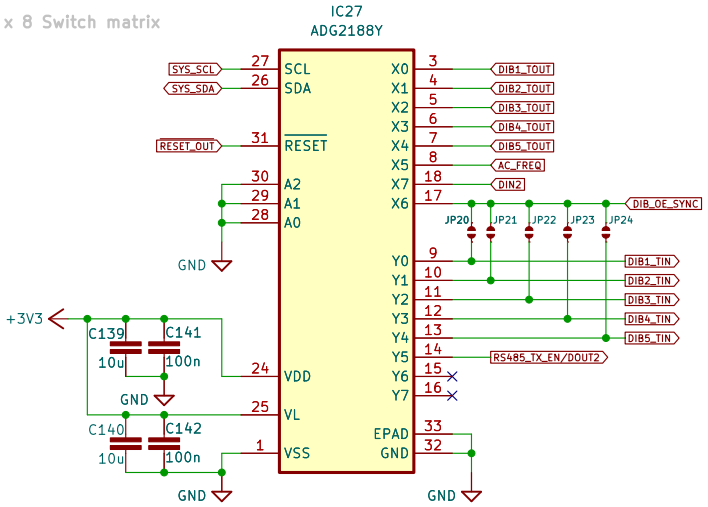
DIB 2.0 Power modules coupling

20-pin DIB v2.0 Power connector

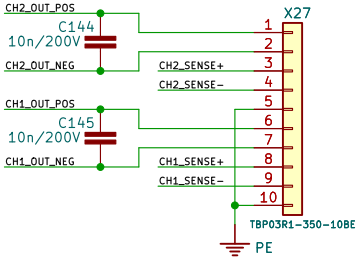
PWR+ IN 1	○○	2	PWR+ IN
PWR+ IN 3	○○	4	PWR+ IN
PWR+ IN 5	○○	6	PWR+ OUT
PWR+ OUT 7	○○	8	PWR+ OUT
PWR+ OUT 9	○○	10	PWR+ OUT
PWR- OUT 11	○○	12	PWR- OUT
PWR- OUT 13	○○	14	PWR- OUT
PWR- IN 15	○○	16	PWR- IN
PWR- IN 17	○○	18	PWR- IN
SENSE- 19	○○	20	SENSE+



Trigger 8 x 8 Switch matrix



Rear front power outputs



Rear panel power connector
(Slots #1 and #2 only)

CH2_OUT+ 1	○
CH2_OUT- 2	○
CH2_SENSE+ 3	○
CH2_SENSE- 4	○
PE 5	○
CH1_OUT+ 6	○
CH1_OUT- 7	○
CH1_SENSE+ 8	○
CH1_SENSE- 9	○
PE 10	○

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.

Sheet: /Trigger switch matrix, Power modules coupling/
File: power_coupling.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-10-31	**Rev: r1B4**
KiCad E.D.A. 10.0.7		Id: 10/10